

Distributions, Regression, and Inference — Comparing Models and Qualifying What They Support

Summarising a distribution, comparing two candidate models through their residuals, and saying exactly how far the resulting inference reaches

Directions:

- **Data Set A** and the two candidate models are in the box below, shared by most of the sheet. Read what you need from there; take everything else from the problem you are on.
- A **residual** is always *observed* – *predicted*. The sign matters: positive means the model under-predicted that point.
- Decide what each problem is asking before computing — some ask you to summarise the data, some to compare the two models, and some to say how far a conclusion can honestly be pushed.
- Round decimal answers to the nearest hundredth unless a problem says otherwise, and give probabilities as exact fractions.

Data Set A — bacterial colony count (hundreds) after x hours

Hour x	1	2	3	4	5	6
Observed count y	6	9	14	21	31	47

Two candidate models fitted to this data**Model L (linear):** $y = 8x - 6$ **Model E (exponential):** $y = 4(1.5)^x$

1. Use **Data Set A**. Find the median of the six observed counts.

Answer: _____

2. Use **Data Set A**. Find the mean of the six observed counts, to the nearest hundredth. Then say which of the mean and the median is larger, and what that ordering tells you about the shape of this distribution.

Answer: _____

3. Model L predicts $y = 8x - 6$. At $x = 3$ the observed count is 14. Find Model L's residual at that hour, and say in words what its sign means.

Answer: _____

4. Model E predicts $y = 4(1.5)^x$. At $x = 3$ the observed count is 14.
Find Model E's residual at that hour.

Answer: _____

5. Compare your two residuals from the previous two problems.
- (a) Which model fits better at that hour, and how do you know from the residuals alone?
 - (b) Explain why a single residual is not enough evidence to choose between two models, and name what you would examine instead.

Answer: _____

- 6.** Use **Data Set A**. Find the interquartile range of the six observed counts. Then say why the interquartile range describes this data differently from the range.

Answer: _____

- 7.** Use Model L, $y = 8x - 6$, at $x = 6$, where the observed count is 47. Find the residual there, and compare its size with the residual you found for Model L at the earlier hour. What does the change tell you about where Model L goes wrong?

Answer: _____

8. Four of Model E's six residuals are positive, and the other two are exactly zero. One of the six hours is selected at random. Find the probability that it is an hour with a positive residual. Give an exact fraction.

Answer: _____

9. Use **Data Set A**. Find the total of the six observed counts.

Answer: _____

10. Model E's residuals, hour by hour, are:

Hour	1	2	3	4	5	6
Residual	0	0	0.5	0.75	0.625	1.4375

At which hour is Model E's residual largest? Then describe the pattern in the signs and sizes of these residuals, and say what that pattern reveals about Model E that a single "it fits well" verdict would hide.

Answer: _____

11. The experiment ran for six hours only.

A lab partner wants to use Model E to state the colony count at hour twelve. Explain what Model E can and cannot support about that claim. Your answer should address the observed range, the behaviour of exponential growth in a real culture, and the fact that this is one run of one experiment.

Answer: _____

12. Challenge. Write the conclusion paragraph of the lab report.

It must: name which model fits Data Set A better and cite residual evidence for that choice; state the one systematic weakness the residual pattern exposes even in the better model; give a prediction together with the range over which you are willing to defend it; and say what a second, independent run of the experiment would add that more arithmetic on this run cannot.

Answer: _____